CLASS 1.2 · INSTRUCTOR NOTES

Soldering Bootcamp

How to teach this class — what happens, how to run it, and why it matters. Read this before you stand at the bench.

THE ONE GOAL

Every kid solders something. It is the one non-negotiable outcome of Phase 1 — everything else this class can flex, this cannot.

HOW THIS CLASS IS TAUGHT — MATE TEACHES, YOU FACILITATE

MATE publishes narrated presentations for the exact kit we own (the TriggerFish ROV Guide). **Let MATE teach the concepts** — assign the presentations below as pre-watch homework — and **you facilitate the hands-on work**. That split is the Section 4.3-friendly pattern and it protects meeting time for the iron. This is a single three-hour session — all hands on irons.

BEFORE YOU TEACH THIS

- **Watch the pre-watch presentations yourself** — MATE's Using Tools and Soldering & Waterproofing Wires. Skim the instructor-only Tools and Tool Management in the Classroom.
- **Make three joints of your own.** Get yours shiny before you demonstrate — you teach by showing, not telling.
- **Read the assigned textbook pages.** *Underwater Robotics* Ch 8 §4.8 *Open Circuits and Short Circuits* (p. 382) and the tools/materials sections (§9.1–9.3, pp. 195–202). The §4.8 reading is the "why" below.
- **Order the SeaMATE kits now** — a Wire Soldering Lab Kit and a PufferFish Practice Board per student (2+ wk lead). Skim the Triggerfish manual's build sequence.
- **Internalize Section 4.3:** demonstrate once, then step back. Never hold the iron for a student — not even to "save time."

WHY THIS CLASS MATTERS

A robot fails at its weakest joint, and that joint is one a kid will make by hand. Soldering is the first real engineering skill of the season: a good connection is the difference between a thruster that runs the whole mission and one that dies in the water, where no one can reach it. Get every kid to a clean, reliable joint and the whole season has a foundation to build on — leave it shaky and you spend the year chasing intermittent faults underwater.

THE ELECTRICAL "WHY" — FROM THE ASSIGNED READING (CH 8 §4.8, P. 382)

Current is simply the flow of charge through a conductor; a solder joint's whole job is to make that path *continuous*. The textbook names the two ways it fails. A cold or loose joint is an **open circuit** — "a gap… preventing current from flowing… a wire breaks or comes loose… a common cause of equipment problems." That is the thruster that quits mid-mission. A solder bridge is a **short circuit** — an "unauthorized shortcut" that bypasses the load and draws "dangerously high current… capable of melting wires… and starting fires… a serious threat to human safety." So **shiny and solid = a closed circuit; dull or bridged = an open or a short.** That is the standard, in the book's own words.

0:00 · 15 MIN Welcome & safety brief

Open with the standing four-question brief, today focused on hot irons — let the CEO run it. Name the hazards out loud: irons at ~700°F, burns, lead and flux fumes, eye spatter. Confirm the fume fan is on and glasses are down before any iron is powered. This brief matters most on the day there's real danger in the room.

0:15 · 15 MIN Why we solder

Set the "why" before the "how," then show the three joints — cold, good, bridged. Use the reading: a good joint is a closed circuit, a cold one is an open (no current), a bridge is a short. Hold up real examples if you have them. A kid who can see and name the difference can fix their own work — that's the whole skill.

0:30 · 25 MIN Recap the pre-watch & the five moves

The concepts were pre-watched at home, so recap quickly, then demonstrate the sequence once: strip, tin the tip, heat the joint, feed the solder, heat-shrink, inspect. Heat the wires, not the solder; feed into the joint until it wicks, then stop. This is the teach-once moment — a clean, unhurried demo sets the standard for every joint that follows.

0:55 · 60 MIN Hands-on practice joints — the heart of the day

This is the block that cannot be cut. Every kid strips, tins, joins, heat-shrinks, and inspects on the SeaMATE practice kits until their joint is steady. Circulate, photograph, encourage — and keep your hands off the irons. Nobody opens the kit until they've made a joint they're proud of, and expect redos: the skill arrives in the redo, not the first try.

THE PARENT LANE — SECTION 4.3

Demonstrate once, then step back. Photograph the build; never hold the iron. MATE's Section 4.3 requires the work to be the students' — including the joints they redo three times. Any outside help, including AI, is disclosed in the documentation. Honesty is scored in April. **An adult hand on the iron is the failure that costs points later.**

MATE RESOURCES FOR THIS CLASS

Assign the first two as pre-watch homework; keep the rest as in-room reference. Source: the TriggerFish ROV Guide.

- Using Tools (20–30 min) PRE-WATCH
- Soldering & Waterproofing Wires PRE-WATCH
- Soldering Wires activity ACTIVITY
- Solder & Seal Connectors REFERENCE
- How to Solder Components to a PCB REFERENCE
- Tools and Tool Management in the Classroom INSTRUCTOR

COMMON PROBLEMS & QUICK FIXES

SYMPTOM	WHAT'S WRONG & THE FIX
Joint is dull, grainy, lumpy	A cold joint — an open circuit waiting to happen. Not enough heat, or it moved while cooling. Reheat until the solder flows bright, then hold dead still until it sets.
Solder won't stick to the wire	Tip isn't tinned, or the wire isn't hot/clean. Re-tin the tip on the sponge, heat the wire first, then feed solder.
Blob bridges two pads	A short circuit — too much solder jumped to the neighbor. Wick the extra off with braid and redo with less.
Iron tip is black and crusty	Oxidized. Wipe on the wet sponge and re-tin with fresh solder until it's shiny again.
Kid is frustrated after two tries	Normal — say so. Pros redo joints constantly. Name the redo as the skill arriving, not failure.

WHAT "DONE" LOOKS LIKE

- Six kids who have each soldered something — at least one clean joint they're proud of.
- Every kid can tell a cold joint from a good one on sight.

PROTECT THE HANDS-ON TIME

Soldering is the one non-negotiable outcome of Phase 1. If the session runs long, protect the practice so every kid leaves with a clean joint, and let anything else slide. The kit comes out next, in Class 1.3.